

Transmission of basal variability to a glacier surface

G. Hilmar Gudmundsson¹

British Antarctic Survey, Cambridge, UK

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[1] Transmission of basal variability to a glacier surface is investigated using analytical models for a linearly viscous medium. The three-dimensional transient response of the surface to both bedrock undulations and spatial variations in basal slipperiness for perturbations of arbitrary wavelengths is determined using perturbation methods. Both information transfer toward the surface and lateral transmission of horizontal stresses are strongly affected by the slip ratio, that is, the ratio of basal sliding to deformational velocity. For any mean bedrock slope, and above a minimum value of slip ratio, the amplitude transfer of bedrock undulations toward the surface has a local maximum at undulation span corresponding to about 3–8 times the mean ice thickness. The transmission of basal variability to a glacier surface increases quite significantly with increasing slip ratio. This explains why the surfaces of fast flowing ice streams are more undulating than the slower moving bordering areas. At slip ratios higher than about 100, the flow of glaciers and ice sheets becomes nonlocal in the sense that surface velocities and buildup and propagation of surface undulations cannot be calculated accurately on the basis of local thickness and slope. Using linearized long-wave theories at these slip ratios, instead of the more accurate arbitrary wavelength theory, gives estimates of decay times that are an order-of-magnitude too small and phase velocities several times too large. The problem of the propagation and decay of small-amplitude surface undulations on glaciers in three dimensions is solved.

Small-amplitude surface waves on glaciers are strongly diffusive and dispersive.

Redistribution of mass on ice sheets and glaciers is a diffusion process, and it is misleading, albeit not mathematically incorrect, to describe the reaction of glaciers to surface perturbations in terms of a wave propagation.

INDEX TERMS: 1827 Hydrology: Glaciology (1863); 3210 Mathematical Geophysics: Modeling; 6020 Planetology: Comets and Small Bodies: Ice; 9310 Information Related to Geographic Region: Antarctica; KEYWORDS: glaciers, ice streams, mathematical geophysics, kinematic waves

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1. Introduction

[2] Most current theories of glacier flow describe the flow of glaciers and ice sheets over spatial scale large compared to the mean ice thickness. Such large-scale models, or shallow ice models, do not describe the effects of spatial changes in either bedrock geometry or mechanical coupling between bed and the basal ice over short scales (on the order of one mean ice thickness). Short-scale basal variability can have strong effects on overall flow velocities and the dynamics of ice sheets. Distributed short scale bedrock undulations, or spatial variations in resistance to basal sliding can, for example, serve as basal stress asperities and are thought to play an important control on the flow of some fast flowing Antarctic ice streams [Alley, 1993]. Understanding how basal variability affects the surface is

also a prerequisite for extracting information about basal conditions from surface data. In this respect it is of particular importance to be able to distinguish between the effect of two limiting examples of basal disturbances on the surface: (1) a bedrock undulation and (2) a spatial variation in basal lubrication along a flat bed.

[3] This paper develops analytical solutions based on perturbation theory describing the transient evolution of flow disturbances in a free-surface gravity-driven linear Stokes flow down an inclined plane in three dimensions. The flow disturbances are set up by both bedrock topography and spatial variations in basal lubrication. Although the applicability of the resulting theory is not limited to ice masses, it will be used here for the purpose of analyzing the transient response of glaciers and ice sheets to basal variability.

[4] The mathematical model takes into account rigorously the effects of horizontal stress gradients. It is therefore not inherently limited to spatial scales which are long in comparison with the mean ice thickness. Basal motion is described using a linear sliding law relating the basal sliding

¹Also at Section of Glaciology, Laboratory of Hydraulic, Hydrology, and Glaciology, ETH-Zurich, Zurich, Switzerland.

velocity to basal shear stress. Rheological nonlinearities are ignored, which is an acceptable approximation as long as the stress perturbations remain small enough for the flow law to be linearized about the mean stress field. The three-dimensional transient response of ice masses to small-amplitude bed and basal slipperiness perturbations is determined for both long and arbitrary wavelengths. Solutions describing the transient evolution of surface undulations having finite transverse widths are given for the first time.

2. Previous Work

[5] The flow of glaciers and ice sheets over bed topography has been an area of active research over the last three decades. Mathematical work has mainly focused on the flow of linear media, and analytical solutions have been found describing transient two dimensional flow [Hutter, 1980, 1983; Jóhannesson, 1992] and steady state flow in three dimensions [Reeh, 1987; Jóhannesson, 1992]. The effects of spatial variations in resistance to basal sliding have been studied by Balise [1987], Hindmarsh [1998a], and Raymond [1996] but, in general, received less attention.

[6] Flow of a linearly viscous medium over small harmonic bedrock undulations was first studied theoretically by Yosida [1964]. Budd [1970b, 1971] attempted to extend Yosida's work to nonlinear flow. One of the earliest theoretical estimates of the effect of bed topography on the glacier surface was by Nye [1959], who considered the transfer of undulations to the surface over length scales which are long compared to the mean ice thickness. Early work on the relationship between surface and bed topography involve the works by Collins [1968] and Budd [1970a].

[7] Analysis of time-dependent changes in the surface geometry of glaciers has traditionally been done using the (linear) kinematic wave equation. The kinematic wave equation was derived by Weertman [1958] and Nye [1960] by assuming the mass flux to be locally dependent on ice thickness and surface slope. It was extensively analyzed by Nye [1963a, 1963b, 1965] and Lliboutry [1971]. An early analysis of time-dependent changes in the surface geometry was done by Finsterwalder [1907], who, based on prevailing knowledge about glacier mechanics, assumed the mass flux to be a local function of thickness only. The effects of stress gradients on the adjustment of a glacier surface to thickness changes was first studied numerically and analytically by Landon and Raymond [1978] and C. F. Raymond (Numerical calculation of glacier flow by finite elements methods, unpublished final technical report for National Science Foundation grant DPP74-19075, 1978) for the no-slip case. Lick [1970] calculated the propagation and decay of small surface disturbances for waves having wavelengths either considerably shorter or longer than the mean ice thickness. Thompson [1979], using a nonlinear flow law for ice, showed that small-amplitude surface waves on glaciers are stable and propagate with a phase velocity which depends on the wavelength. The phase velocity of surface waves has also been studied numerically by van de Wal and Oerlemans [1995].

[8] The first systematic analysis of the effects of basal undulations on glacier flow using perturbation methods was done by Hutter [1980, 1982, 1983] and Hutter et al. [1981].

Hutter's work, which was limited to two dimensions, was extended by Reeh [1987] for the linear steady state problem in three dimensions in the absence of basal motion. Whillans and Johnsen [1983] considered the two dimensional problem for an exponentially varying viscosity distribution with depth. They investigated both flow disturbances due to bedrock undulations and a prescribed velocity variation along the lower boundary. Balise and Raymond [1985] analyzed the initial response of a linear medium to basal velocity perturbations in two dimensions, and Balise [1987] investigated the effects of a spatially varying sliding law parameter on the flow field of both a linear and nonlinear medium. Jóhannesson [1992] reanalyzed, corrected and significantly extended previously obtained results for two and three dimensions. He derived solutions for the transient two dimensional problem, and solved the three-dimensional steady state problem with basal sliding. Raymond [1996] investigated the effects of transverse variations in lubrication of the bed on the flow of ice sheets and used the resulting theory to analyze the mechanics of flow around shear margins.

3. Nonlinear Long Waves

[9] A valuable first insight into the response of ice masses to basal disturbances can be gained from long-wavelength analysis. This type of analysis is based on the assumption that the shallowness perturbation parameter $\delta = d/l$ is small, where d is the slab thickness and l is the horizontal distance scale. Most numerical models of ice sheets are based on the simplifying assumption $\delta \rightarrow 0$. The prime example of this is the shallow ice approximation [Hutter, 1983], which is a zeroth-order long-wavelength model. Here the linearized version of the zeroth-order long-wavelength problem in a tilted coordinate system is considered for a linearly viscous medium. This analysis is used to obtain analytical solutions describing the transient surface response to bedrock and basal slipperiness perturbations in three dimensions. Because of the role that the $\delta \rightarrow 0$ approximation currently plays in glacier mechanics, it is particularly interesting to compare the results of this analysis with the subsequent more general analysis done for arbitrary waves in section 4. In the long-wavelength limit, the flow velocity is related to local slope and thickness only, and horizontal deviatoric stresses are not transmitted. An overview of the use of this long-wavelength approximation in glaciological modeling, and discussion of other related issues is given by Fowler [2000].

3.1. Finite Amplitude Long Waves

[10] Following Fowler and Larson [1980] and Fowler [1982], the scalings employed are

$$(x, y) = l(X, Y), \quad z = dZ, \quad (v_x, v_y) = u_d(V_X, V_Y), \quad (1)$$

$$v_z = \delta u_d V_Z, \quad t = l u_d^{-1} T, \quad p = P \tau_b \cot \alpha, \quad (2)$$

where ρ is the ice density, u_d is the typical contribution of internal deformation to the surface velocity ($\tau_b d / 2\eta$), τ_b is a typical basal shear stress, η the ice viscosity, and l a typical horizontal length scale. The mean surface slope is α with

$\sin \alpha = \sqrt{\sin^2 \alpha_X + \sin^2 \alpha_Y}$, where $\sin \alpha_X = \hat{\mathbf{e}}_X \cdot \hat{\mathbf{g}}$, $\sin \alpha_Y = \hat{\mathbf{e}}_Y \cdot \hat{\mathbf{g}}$, with $\hat{\mathbf{e}}_i$ being the unit coordinate vectors for $i = x, y, z$. The coordinate system is tilted by the angles α_X and α_Y in x and y direction, respectively. Nondimensional variables are generally denoted by upper case letters and dimensional ones by lower case letters, but there are exceptions to this rule. All variables are defined in the text and listed in the notation section.

[11] The momentum equation is

$$p_{,i} = \eta \nabla^2 v_i + f_i, \quad (3)$$

where f_i for $i = 1, 2, 3$ are the components of the body force per unit volume, p is the pressure, and v_i are the components of the velocity vector. The incompressibility equation is

$$v_{i,i} = 0. \quad (4)$$

These scalings lead to

$$\mu \partial_X P = \frac{1}{2} (\delta^2 \partial_X^2 V_X + \delta^2 \partial_Y^2 V_X + \partial_Z^2 V_X) - \frac{\sin \alpha_X}{\sin \alpha} \quad (5)$$

$$\mu \partial_Y P = \frac{1}{2} (\delta^2 \partial_X^2 V_Y + \delta^2 \partial_Y^2 V_Y + \partial_Z^2 V_Y) - \frac{\sin \alpha_Y}{\sin \alpha} \quad (6)$$

$$\cot \alpha \partial_Z P = \frac{1}{2} (\delta^3 \partial_X^2 V_Z + \delta^3 \partial_Y^2 V_Z + \delta \partial_Z^2 V_Z) - \cot \alpha \quad (7)$$

where $\mu := \delta \cot \alpha$, and capital letters denote corresponding scaled variables.

[12] The surface position is described by the function $s(x, y, t)$. Ignoring surface accumulation and ablation, the kinematic boundary condition for the material surface reads

$$\partial_t s + v_x \partial_x s + v_y \partial_y s = v_z \quad z = s(x, y, t), \quad (8)$$

The jump condition for the stress tensor gives

$$\sigma_{ij} m_j = 0 \quad z = s, \quad (9)$$

where $\mathbf{m}$ is a unit outward normal vector to the surface and σ_{ij} are the components of the Cauchy stress tensor.

[13] The flow fields fulfilling the field equations are quasi steady since the time variable t does not appear explicitly in equations (3) and (4). It is the unsteady evolution of the free upper surface, which gives rise to time-dependent flows.

[14] The vertical elevation of the lower boundary is given by $b(x, y, t)$. Ignoring basal melting or refreezing implies

$$v_i n_i = 0 \quad z = b(x, y, t) \quad (10)$$

where $\mathbf{n}$ is the outward unit normal vector to $b(x, y, t)$.

[15] A sliding law of the form

$$u_b(x, y, t) = c(x, y, t) \tau_b(x, y, t), \quad (11)$$

is assumed, where u_b is the basal sliding velocity, c a sliding law parameter, and τ_b the basal shear stress. The form of the

sliding law for real glaciers may be more complex than equation (11) indicates. A sliding law of this type is a correct basal condition for the bulk flow at a smoothed bed form corresponding to a variety of physical conditions along the true ice-substrate interface (for example free-slip, no-slip, deforming till) [e.g., Fowler, 1979; Hutter, 1983]. By allowing the sliding law parameter c to be a function of x and y the effects of spatial variations in subglacial conditions (till thickness, basal water pressure etc.) on the sliding process are simulated. In this way flow around “slippery” or “sticky” spots may be analyzed.

[16] Only the form for $\delta \rightarrow 0$ will be considered, and in that limit, the scaled boundary conditions at the upper surface are $\Sigma_{XZ} = \Sigma_{YZ} = 0$, where Σ_{ij} are the scaled components of the Cauchy stress tensor. The kinematic boundary condition (equation (8)) is invariant under the scalings (1). At the lower boundary, the zeroth-order sliding law reads $V_X = C \Sigma_{XZ}$, $V_Y = C \Sigma_{YZ}$, with C scaled using $c = 2\eta C/d^{(0)}$, where $d^{(0)}$ is the mean thickness. The parameter μ will be kept $O(1)$ and balanced with the zeroth-order terms in δ . This balance is crucial if the important effects of diffusion on surface evolution of glaciers and ice sheets are to be correctly described for small surface slopes.

[17] Equation (7) for $\delta \rightarrow 0$ yields $P = S - Z$, and from equation (5) one finds

$$\partial_Z V_i = 2 \left(\frac{\sin \alpha_i}{\sin \alpha} - \mu \partial_i S \right) (S - Z)$$

for $i = X, Y$. The horizontal velocity components are given by

$$V_i = \left(\frac{\sin \alpha_i}{\sin \alpha} - \mu \partial_i S \right) (D^2 - (S - Z)^2 + CD) \quad (12)$$

for $i = X, Y$, where $D = S - B$ is the slab thickness. Vertically integrating the incompressibility equation (4) over the glacier thickness to determine V_Z , and inserting the result into the kinematic boundary condition at the upper surface leads to

$$\partial_t S + \sum_{i \in X, Y} \partial_i \left\{ \left(\frac{\sin \alpha_i}{\sin \alpha} - \mu \partial_i S \right) \left(\frac{2}{3} D^3 + CD^2 \right) \right\} = 0. \quad (13)$$

This is a nonlinear advection-diffusion equation. A corresponding equation for a nonlinear rheology for $D = 1 + S$ and C constant was derived and analyzed in detail by Fowler and Larson [1978, 1980] and Fowler [1982].

3.2. Linearizing

[18] Aligning the coordinate system in the direction of maximum surface slope by setting $\alpha_X = \alpha$ and $\alpha_Y = 0$, and linearizing equation (13) with respect to perturbations in the surface, the bedrock, and the sliding coefficient, gives to first order

$$\begin{aligned} \partial_t S^{(1)} + 2 \left(1 + C^{(0)} \right) \partial_X S^{(1)} - \mu \left(2/3 + C^{(0)} \right) (\partial_X^2 + \partial_Y^2) S^{(1)} \\ = 2 \left(1 + C^{(0)} \right) \partial_X B^{(1)} - C^{(0)} \partial_X C^{(1)} \end{aligned} \quad (14)$$

where $S = S^{(0)} + S^{(1)}$, $B = B^{(0)} + B^{(1)}$, and $C = C^{(0)}(1 + C^{(1)})$. This equation describes small amplitude and long wavelength disturbances. Rescaling equation (14) using the mean thickness alone as the relevant length scale yields

$$\begin{aligned} \partial_t S^{(1)} + 2(1 + C^{(0)}) \partial_X S^{(1)} - (2/3 + C^{(0)}) \cot \alpha (\partial_X^2 + \partial_Y^2) S^{(1)} \\ = 2(1 + C^{(0)}) \partial_X B^{(1)} - C^{(0)} \partial_X C^{(1)}. \end{aligned} \quad (15)$$

This equation will be called the linearized long-wavelength equation. Equation (15) can be compared directly with the linear version of Nye's [1960] original kinematic wave equation. The left-hand side of equation (15) (without the ∂_Y^2 term) is identical to Nye's equation describing the transient evolution of surface disturbances. With the first source term on the right-hand side, equation (15) was derived for the first time by Jóhannesson [1992, equation (5.4.4)].

[19] Equation (15) is a linear advection-diffusion equation with two source terms, and can be solved using standard Fourier and Laplace transforms methods. It is convenient to present the analytical solutions in the form of transfer functions. Long-wavelength transfer functions are denoted by the letter $\bar{T}$. Later transfer functions for arbitrary wavelengths will be derived and denoted by $\mathcal{T}$. The transfer functions have a two letter suffix. The first suffix relates to the effect and the second one to the cause. $\bar{T}_{SB}$, for example, is the transfer function that describes a change in the surface topography ($S^{(1)}$) caused by a bedrock topography ($B^{(1)}$), whereas $\bar{T}_{SC}$ is the transfer function describing a change in surface topography ($S^{(1)}$) caused by a spatial variation in the basal sliding parameter ($C^{(1)}$).

[20] The two-dimensional forward Fourier transform $f(k_x, k_y)$ of a function $f(x, y)$ is

$$f(k_x, k_y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) e^{i(k_x x + k_y y)} dx dy, \quad (16)$$

where i is the imaginary unit. The forward Laplace transform $f(s)$ of a function $f(t)$ is

$$f(s) = \int_{0^+}^{+\infty} f(t) e^{-st} dt. \quad (17)$$

Fourier and Laplace transform of equation (15) followed by an inverse Laplace transform using the methods of contour integration leads to

$$\begin{aligned} S^{(1)}(k_x, k_y, t) = \bar{T}_{SB} B^{(1)}(k_x, k_y) + \bar{T}_{SC} C^{(1)}(k_x, k_y) \\ + \bar{T}_{SS} S^{(1)}(k_x, k_y, t = 0) \end{aligned} \quad (18)$$

where $\bar{T}_{SB}$, $\bar{T}_{SC}$, and $\bar{T}_{SS}$ are long-wavelength transfer functions given by

$$\bar{T}_{SB} = \frac{i}{i - \bar{t}_p / \bar{t}_d} \left(1 - e^{i \bar{t}_p / \bar{t}_d} e^{-t / \bar{t}_d} \right), \quad (19)$$

$$\bar{T}_{SC} = -\frac{i C^{(0)}}{2(i - \bar{t}_p / \bar{t}_d)(1 + C^{(0)})} \left(1 - e^{i \bar{t}_p / \bar{t}_d} e^{-t / \bar{t}_d} \right), \quad (20)$$

$$\bar{T}_{SS} = e^{i \bar{t}_p / \bar{t}_d} e^{-t / \bar{t}_d}, \quad (21)$$

where the two timescales $\bar{t}_p$ and $\bar{t}_d$ are given by

$$\bar{t}_d^{-1} := \left(2/3 + C^{(0)} \right) k^2 \cot \alpha, \quad (22)$$

and

$$\bar{t}_p^{-1} := 2(1 + C^{(0)}) k_x. \quad (23)$$

[21] Equations (19) to (21) show that $\bar{t}_d$ controls decay and risetimes, while the rate of propagation is determined by $\bar{t}_p$; $\bar{t}_d$ is the long-wavelength decay timescale, and $\bar{t}_p$ the long-wavelength propagation timescale. The ratio or $\bar{t}_d / \bar{t}_p$ can be written as

$$\frac{\bar{t}_d}{\bar{t}_p} = \frac{\lambda}{\lambda^*(\theta)} \quad (24)$$

with

$$\lambda^*(\theta) := \frac{\pi (2/3 + C^{(0)}) \cot \alpha}{(1 + C^{(0)}) \cos \theta} \quad (25)$$

and $\cos \theta = k_x / k$ and $\sin \theta = k_y / k$. The wavelength λ is related to the longitudinal wavelength λ_x and the transverse wavelength λ_y through $\lambda^{-1} = \sqrt{\lambda_x^{-2} + \lambda_y^{-2}}$ with $2\pi = k_x \lambda_x = k_y \lambda_y = \lambda k$. The angle θ is the angle between the positive x axis and the vector lying in the xy plane and normal to the crests of the sinusoidal bed perturbations. The ratio λ / λ^* determines the relative importance of the advection and the diffusion terms in equation (15).

[22] For typical surface slopes on glaciers and ice sheets (0.01–0.05), one finds that $\lambda^* \gg 1$. For wavelengths less than about 50–200, the long-wavelength decay time ($\bar{t}_d$) is shorter than the propagation time, as can be seen from equation (24). The amplitude of such a surface wave will have decayed significantly in the time it takes for the wave to travel downstream one wavelength. For steep slopes (> 0.15) and wavelengths larger than about 20, the situation is somewhat different. The situation can be different for short wavelengths where the long-wavelength theory breaks down (see section 5.5).

[23] The ratio between the amplitude of a sinusoidal surface and bedrock undulation follows from equation (18) and is given by

$$|\bar{T}_{SB}| = \frac{\lambda / \lambda^*}{\sqrt{1 + (\lambda / \lambda^*)^2}} \left\{ 1 + e^{-t / \bar{t}_d} \left[e^{t / \bar{t}_d} - 2 \cos(t / \bar{t}_p) \right] \right\}. \quad (26)$$

For the C perturbation the amplitude ratio is

$$|\bar{T}_{SC}| = \frac{C^{(0)}}{2(1 + C^{(0)})} |\bar{T}_{SB}| \quad (27)$$

The phase shift between the surface topography and the basal perturbations can be described by a phase shift function, $\bar{\phi}$. The phase shift of the surface topography with respect to the bedrock is

$$\tan \bar{\phi}_{SB} = \frac{-\bar{t}_d^{-1} + e^{-t/\bar{t}_d} \left(\bar{t}_d^{-1} \cos(t/\bar{t}_p) - \bar{t}_p^{-1} \sin(t/\bar{t}_p) \right)}{\bar{t}_p^{-1} - e^{-t/\bar{t}_d} \left(\bar{t}_p^{-1} \cos(t/\bar{t}_p) + \bar{t}_d^{-1} \sin(t/\bar{t}_p) \right)}, \quad (28)$$

and with respect to a sliding perturbation, the phase shift is $\bar{\phi}_{SC} = \pi + \bar{\phi}_{SB}$.

[24] The steady state long-wavelength amplitude response of both bedrock and sliding perturbations increase monotonically toward unity as functions of λ , with λ^* controlling the rate of increase. For $\theta \rightarrow \pm\pi/2$ the wavelength λ^* becomes infinitely long (equation (25)). Transverse bedrock undulations, thus, do not influence the surface shape and only the forward velocity component is perturbed. For θ to $\pm\pi/2$, there is a perfect transmission of bedrock undulations to the surface in the limit $\lambda \rightarrow +\infty$ (equation (26)).

[25] The steady state infinite-wavelength response of the surface topography to a B perturbation can be determined by direct inspection of the differential equation (15). The equation shows that the diffusion terms go faster to zero with increasing wavelength than the advection term. As a consequence, the perturbations in surface and bedrock slopes go to zero as $\lambda \rightarrow \infty$ for $|\theta| < \pi/2$. The coefficients multiplying $\partial_x S^{(1)}$ and $\partial_x B^{(1)}$ in equation (15) represent the partial derivatives of the horizontal mass flux vector with respect to perturbations in the surface and bedrock geometry, respectively. These partial derivatives must be of same magnitude, but with opposite signs, because the mass flux is increased equally by an upward shift in the position of the surface as by the same downward shift in the position of the bedrock. In the limit $\lambda \rightarrow \infty$, for λ^* kept fixed, the surface is thus in phase with the bedrock and the amplitude ratio between the surface and the bedrock undulations equal to unity, independently of the rheological properties of the medium or the type of the basal sliding law. An identical argument shows that in the same limit, the surface response to a C perturbation is π radians out of phase with the basal perturbation and reaches an amplitude of $C^{(0)}/(2(1 + C^{(0)}))$ in the limit $\lambda \rightarrow +\infty$.

[26] The long-wavelength analysis of spatial sliding variations given above is new, as are the expressions for the time evolution of the surface (equation (21)), and its response to a bedrock perturbation (equations (19) and (20)). Using a fundamentally different mathematical approach, in line with the original work by *Finsterwalder* [1907] and *Nye* [1960], *Jóhannesson* [1992] solved the bedrock-to-surface transient problem, and equation (19) can be derived from his work.

4. Linearized Waves of Arbitrary Wavelengths

4.1. Problem Definition

[27] The field and the boundary equations (equations (3), (4), (8), (9), and (10)) are now scaled with the mean thickness in all three spatial directions. The undisturbed flow, the zeroth-order solution, is described by the plane-slab solution. Analytical solutions are found by linearizing about this

ambient state. As above, rectangular Cartesian coordinates are used with the x axis along the upper surface in the direction of the zeroth-order flow, the y axis horizontal across the slope, and the z axis normal to the zeroth-order surface. The slab has a mean thickness $d^{(0)}$ and a mean slope α .

4.2. Nondimensionalization

[28] All dimensional variables entering the problem are substituted by nondimensional counterparts through the use of scalings suggested by the properties of the zeroth-order solution. In general the nondimensional, or scaled, variables are written in uppercase letters that correspond to the lowercase letters of the dimensional variables.

[29] The velocities are scaled by the deformational velocity of the zeroth-order solution $u_d^{(0)}$, the stresses and the pressure by the zeroth-order basal shear stress $\tau_b^{(0)}$, and the spatial dimensions by the zeroth-order ice thickness $d^{(0)}$, where the superscript refers to the order of the solution. The deformational velocity of the zeroth-order solution is $u_d^{(0)} = \frac{1}{2\eta} \tau_b^{(0)} d^{(0)}$, and the zeroth-order basal shear stress is $\tau_b^{(0)} = \rho g d^{(0)} \sin \alpha$, where $d^{(0)} = s^{(0)} - b^{(0)}$, and $s^{(0)}$ and $b^{(0)}$ are the upper and the lower boundaries of the zeroth-order solution, respectively. Summarizing, the scalings are $v_i = u_d^{(0)} V_i$, $(p, \sigma_{ij}) = \tau_b^{(0)} (P, \Sigma_{ij})$, and $(x, y, z) = d^{(0)} (X, Y, Z)$, where V_i are the scaled components of the velocity vector, P is the scaled pressure, Σ_{ij} are the scaled components of the Cauchy stress tensor, and X, Y and Z are the scaled spatial variables. These scalings imply a timescale $T_s = d^{(0)}/u_d^{(0)} = 2\eta/\tau_b^{(0)}$ if the form of the field equations, and in particular the form of the kinematic boundary condition at the surface, is to be the same in nondimensional as in dimensional variables.

[30] The spatial variables are scaled equally in all three spatial directions. This represents the main difference with respect to the shallow ice approximation [*Morland and Johnson*, 1980; *Hutter*, 1983, 1993], where x and y are scaled with a length scale large comparable to an ice sheet span. The treatment of bed topography in the shallow ice approximation and ways of improvement are discussed by *Morland* [2000]. Here the focus is on a limited area of an ice sheet. The formulation is done in a tilted coordinate system, and the analysis deals with small perturbations in the thickness of a parallel slab. The shallow ice approximation, and the “enhanced reduced model” of *Morland* [2000], are formulated with respect to horizontal and vertical coordinates and deal with large-scale ice sheet dynamics.

[31] The set of scalings defined above is referred to as short-scale scalings. In nondimensional variables the boundary conditions are simply equations (8), (9), (10), and (11) in capital letters, with $c(x, y, t)$ replaced by

$$C(X, Y, T) = \frac{2\eta}{d^{(0)}} c(x, y, t). \quad (29)$$

4.3. Formulation of the Perturbation Approach

[32] The B perturbation is introduced by writing the function $B(X, Y, T)$ describing the basal topography as

$$B(T, X, Y; \varepsilon) = B^{(0)} + \varepsilon F_b^{(\varepsilon)}(T, X, Y), \quad (30)$$

where $B^{(0)}$ stands for the mean bedrock elevation and $F_b^{(\varepsilon)}(X, Y, T)$ the variation around this mean value. The ratio

and

$$\varepsilon V_Y^{(\varepsilon)} + \beta V_Y^{(\beta)} = C^{(0)} \left(\varepsilon \Sigma_{YZ}^{(\varepsilon)} + \beta \Sigma_{YZ}^{(\beta)} \right), \quad (45)$$

[42] The boundary conditions (39), (40), (41), (42), (43), (44), and (45) form a set of seven first-order boundary conditions in ε and β for the B and the C perturbation problems.

4.6. First-Order Boundary Conditions of the S Perturbation Problem

[43] The S perturbation represents an initial condition rather than a boundary condition and can be thought of as an initial condition for either the C or the B perturbation problems. The S perturbation problem does therefore not have to be considered separately. To be complete I nevertheless list the first-order boundary conditions of the S perturbation problem, which are

$$\partial_T F_s^{(\varepsilon)} + V_s^{(0)} \partial_X F_s^{(\varepsilon)} = V_Z^{(\varepsilon)} \quad (46)$$

$$\Sigma_{XZ}^{(\varepsilon)} = F_s^{(\varepsilon)} \quad (47)$$

$$\Sigma_{YZ}^{(\varepsilon)} = 0, \quad (48)$$

$$\cot(\alpha) F_s^{(\varepsilon)} + \Sigma_{ZZ}^{(\varepsilon)} = 0, \quad (49)$$

at $Z = 0$, and

$$V_Z^{(\varepsilon)} = 0 \quad (50)$$

$$V_X^{(\varepsilon)} = C^{(0)} \Sigma_{XZ}^{(\varepsilon)}, \quad (51)$$

$$V_Y^{(\varepsilon)} = C^{(0)} \Sigma_{YZ}^{(\varepsilon)}, \quad (52)$$

at $Z = -1$.

4.7. Transformed First-Order Field Equations

[44] The first-order differential equations derived from equations (3) and (4) form a system of linear partial differential equations that can be solved using standard Fourier and Laplace transform methods. In the following the expressions are written using dimensional variables as it is easier to go from the dimensional to the nondimensional form.

[45] Transforming the momentum and the mass equation (3) and (4) using the definitions of the Fourier and the Laplace transforms (equations (16) and (17)) leads to a fourth-order differential equation for v_z

$$\partial_z^4 v_z^{(1)} - 2k^2 \partial_z^2 v_z^{(1)} + k^4 v_z^{(1)} = 0, \quad (53)$$

with the pressure distribution determined by

$$p^{(1)} = \eta \left(\frac{1}{k^2} \partial_z^3 v_z - \partial_z v_z \right), \quad (54)$$

where $k := \sqrt{k_x^2 + k_y^2}$ denotes the modulus of the wave number vector. Whenever the superscript 1 is used, it is to be understood that it can be substituted by any one of the three perturbation parameters ε , β , or $\hat{\varepsilon}$.

[46] The solution to equation (53) is

$$v_z^{(1)} = (A_h + B_h k z) \sinh(kz) + (C_h + D_h k z) \cosh(kz) \quad (55)$$

where A_h , B_h , C_h and D_h are unknown constants. The pressure field can now be determined using equation (54), giving

$$p^{(1)} = 2k\eta [B_h \sinh(kz) + D_h \cosh(kz)]. \quad (56)$$

With p and v_z known, one then finds

$$v_x^{(1)} = \iota \left\{ \left(-E_h \frac{k}{k_x} - D_h k z + B_h \frac{k_x}{k} \right) \sinh(kz) + \left(-F_h \frac{k}{k_x} - B_h k_x z + D_h \frac{k_x}{k} \right) \cosh(kz) \right\}, \quad (57)$$

$$v_y^{(1)} = \iota \left\{ \left(-G_h \frac{k}{k_y} - D_h k_y z + B_h \frac{k_y}{k} \right) \sinh(kz) + \left(-H_h \frac{k}{k_y} - B_h k_y z + D_h \frac{k_y}{k} \right) \cosh(kz) \right\}, \quad (58)$$

with

$$E_h + G_h = 2B_h + C_h \quad ; \quad F_h + H_h = 2D_h + A_h, \quad (59)$$

where E_h , F_h , G_h and H_h are determined by the boundary conditions. The solutions listed above are in dimensional form. To arrive at the nondimensional form all field variables must be changed to upper case letters and the viscosity η replaced by the numerical factor $1/2$. The values for the variables A_h to G_h are determined by the boundary conditions.

4.8. Transformed First-Order Boundary Conditions

[47] The external perturbations are applied at $t = 0$. Taking the Fourier and the Laplace transform of the non-dimensionalized first-order kinematic boundary condition (39) leads to

$$S^{(1)} = \frac{V_Z^{(1)} + S^{(1)}(t = 0^+)}{s - \iota k_x V_s^{(0)}}. \quad (60)$$

The term $S^{(1)}(t = 0^+, k_x, k_y)$ represents the S perturbation which is an initial condition.

[48] It turns out that expression (60) can be used to eliminate $S^{(1)}(s, k_x, k_y)$ from the two jump conditions (40) and (41) leaving

$$\Sigma_{XZ}^{(1)} - \frac{V_Z^{(1)} + S^{(1)}(t = 0^+, k_x, k_y)}{s - \iota k_x V_s^{(0)}} = 0, \quad (61)$$

$$\Sigma_{ZZ}^{(1)} + \cot \alpha \frac{V_Z^{(1)} + S^{(1)}(T = 0^+)}{s - \iota k_x V_s^{(0)}} = 0, \quad (62)$$

$$\Sigma_{YZ}^{(1)} = 0, \quad (63)$$

as the three boundary conditions at the upper surface. Along the lower boundary the no-penetration condition and the sliding law take the forms

$$V_Z^{(1)} = -\iota \frac{k}{s} C^{(0)} B^{(1)}(k_x, k_y), \quad (64)$$

$$V_Y^{(1)} - C^{(0)} \Sigma_{YZ}^{(1)} = 0, \quad (65)$$

$$V_X^{(1)} - C^{(0)} \Sigma_{XZ}^{(1)} = \frac{1}{s} \left\{ -\left(2 + C^{(0)}\right) B^{(1)}(k_x, k_y) + C^{(0)} C^{(1)}(k_x, k_y) \right\} \quad (66)$$

Note that

$$C^{(1)}(k_x, k_y) := \frac{G^{(1)}(k_x, k_y)}{C^{(0)}}, \quad (67)$$

and

$$B^{(1)}(k_x, k_y) := \varepsilon F_b^{(\varepsilon)}(k_x, k_y), \quad (68)$$

so that in nondimensional form the sliding law is

$$U_b = C^{(0)} \left(1 + C^{(1)}(k_x, k_y)\right) \Sigma_b, \quad (69)$$

where Σ_b is the basal shear traction.

[49] There are seven boundary conditions that enter the problem, four at the upper boundary, and three at the lower boundary. Since the transformed kinematic boundary condition (60) was used to eliminate the $S^{(1)}(s, k_x, k_y)$ field from the other three surface conditions, the remaining six boundary conditions (equations (61) to (66)) can be used together with equations (55) to (58) and the corresponding expressions for the stresses (not listed) to determine the constants A_h to H_h . Equation (60) then gives the time evolution of the surface.

4.9. Solutions

[50] The transfer functions describing the effects of the S and B basal perturbations are found by solving the set of the transformed boundary equations equations (61) to (66) together with the corresponding transformed field equations. A few examples for the resulting transfer functions are given by

$$\mathcal{T}_{SB}(s, k_x, k_y) = \frac{-\iota \tilde{a}/\tilde{c}}{s[s - (\iota/t_p - 1/t_d)]}, \quad (70)$$

and

$$\mathcal{T}_{SC}(s, k_x, k_y) = \frac{-\iota \tilde{e}/\tilde{c}}{s[s - (\iota/t_p - 1/t_d)]}, \quad (71)$$

which describe the transient response of the surface to a B and a C perturbation, respectively. In equations (70) and (71),

$$t_d := \frac{\tilde{c}}{b}, \quad (72)$$

$$t_p := \frac{\tilde{c}}{d}, \quad (73)$$

and

$$\tilde{a} := \left\{ V_s^{(0)} \tilde{f} + \left(V_s^{(0)} + k^2 C^{(0)^2} \right) \cosh(k) \right\} k k_x,$$

$$\tilde{b} := \left\{ \tilde{f} \sinh(k) - k \right\} \cot \alpha,$$

$$\tilde{c} := k^3 V_s^{(0)} + \tilde{f} k \cosh(k) \quad , \quad \tilde{d} := k_x V_s^{(0)} (\tilde{c} + k),$$

$$\tilde{e} := -k_x k C^{(0)} \cosh(k) \quad , \quad \tilde{f} := \cosh k + k C^{(0)} \sinh k.$$

$\mathcal{T}_{SB}(s, k_x, k_y)$ and $\mathcal{T}_{SC}(s, k_x, k_y)$ go to zero as $s \rightarrow +\infty$, which is a necessary condition for them to be well defined. The time-dependent transfer functions t are calculated using the Bromwich integral

$$F(t) = \frac{1}{2\pi\iota} \int_{\gamma-\iota\infty}^{\gamma+\iota\infty} e^{st} f(s) ds, \quad (74)$$

which is evaluated by contour integration. Because their denominators are identical, the transfer functions $\mathcal{T}_{SB}(s, k_x, k_y)$ and $\mathcal{T}_{SC}(s, k_x, k_y)$ share the same poles. This is a general property of all the C and B transfer functions, and follows from the structure of the system of equations (61) to (66). For this reason the timescales entering the three perturbations problems are all identical. The function t_d is nonnegative for all values of k and the solutions, therefore, always stable.

[51] For $\gamma > 0$ both singularities of the transfer functions are to the left of the line of integration. Inverting expressions (70) and (71) by contour integration for $t > 0$, where the contour is closed by an infinite semi circle in the left half plane, one obtains

$$\mathcal{T}_{SB}(t, k_x, k_y) = \frac{\iota \tilde{a}/\tilde{c}}{\iota t_p^{-1} - t_d^{-1}} \left(1 - e^{\iota t/t_p} e^{-t/t_d}\right), \quad (75)$$

$$\mathcal{T}_{SC}(t, k_x, k_y) = \frac{\iota \tilde{e}/\tilde{c}}{\iota t_p^{-1} - t_d^{-1}} \left(1 - e^{\iota t/t_p} e^{-t/t_d}\right). \quad (76)$$

These functions describe the transient response of the surface to B and C perturbations applied at $t = 0$. For $t = 0$, both $\mathcal{T}_{SB}$ and $\mathcal{T}_{SC}$ are equal to zero because it takes a finite period of time for any surface disturbances to be created from an initially flat surface through viscous flow. In the steady state limit $t \rightarrow +\infty$, the transfer of the B and the C basal perturbations to the surface are given by the ratios $\tilde{a}/(\tilde{d} + \iota \tilde{b})$ and $\tilde{e}/(\tilde{d} + \iota \tilde{b})$, respectively.

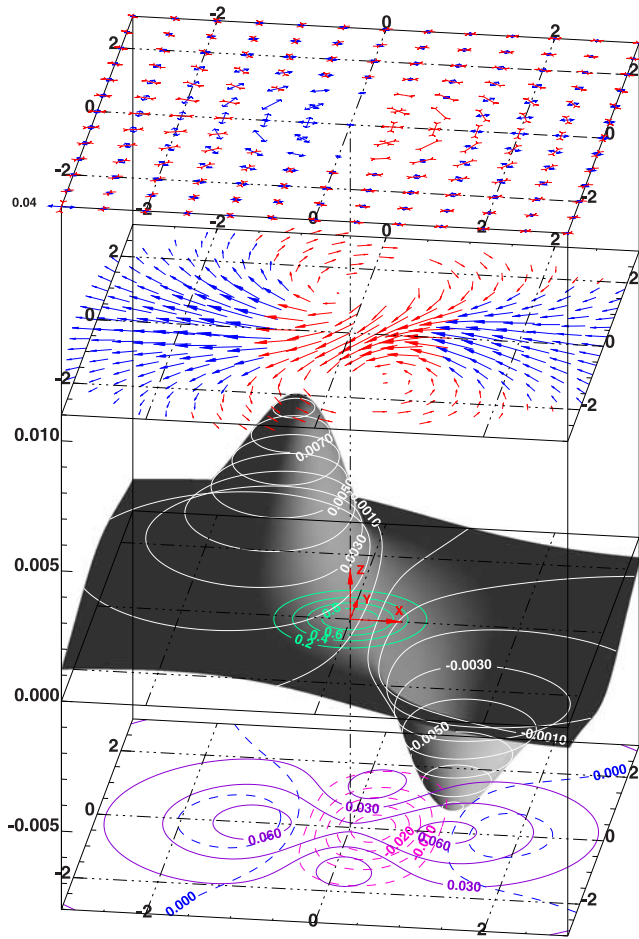


Figure 2. Same as in Figure 1 but for a Gaussian-shaped negative perturbation in basal slipperiness.

[58] Some very general features of the surface response to both types of basal perturbations are similar (Figures 1 and 2). For example, in both cases the ice piles up on the upstream side of the center of the basal disturbances while a surface depression is formed on the down stream side. Most other aspects are markedly different. In Figure 1 the direction of the perturbed flow changes twice along the medial line ($y = 0$), while in Figure 2 the perturbed flow is always opposite to the mean flow field. In Figure 2 the surface response is broader and the surface amplitudes smaller than in Figure 1. The strain rate fields (Figures 1 and 2, top plane) are complex, with alternating regions of horizontal extension (blue strain rate arrows) and compression (red strain rate arrows). In general, the surface flow field induced by the slipperiness perturbation (Figure 2) is around the basal perturbation and to a lesser extend directly across it as compared to Figure 1. Horizontal speeds (purple contour lines along the lowest planes in Figures 1 and 2) are also markedly different with five local maximum points in Figure 1 all located along the medial line, and four in Figure 2 of which only two are located on medial line.

[59] These significant differences in flow characteristics suggest that it may be possible to distinguish between these two types of basal perturbations from surface data. In fact, it can be shown that knowledge of surface topography and at least one surface velocity component is sufficient for

inverting simultaneously for both slipperiness and bed profile [Gudmundsson *et al.*, 2001; Thorsteinsson *et al.*, 2003].

5.2. Amplitude Ratios and Phase Shifts

[60] The properties of the above listed complex transfer functions can be best understood by analyzing them in terms of the amplitude ratios and phase shifts between surface and basal fields. The amplitude ratios of surface undulations caused by either harmonic B and C perturbations follow from equations (75) and (76) and are given by

$$|\mathcal{T}_{SB}(t, k_x, k_y)| = \frac{\tilde{a}/\tilde{c}}{\sqrt{t_p^{-2} + t_d^{-2}}} \left\{ 1 + e^{-t/t_d} \left(e^{-t/t_d} - 2 \cos(t/t_p) \right) \right\}^{1/2}, \quad (82)$$

and

$$|\mathcal{T}_{SC}(t, k_x, k_y)| = \frac{-\tilde{e}/\tilde{c}}{\sqrt{t_p^{-2} + t_d^{-2}}} \left\{ 1 + e^{-t/t_d} \left(e^{-t/t_d} - 2 \cos(t/t_p) \right) \right\}^{1/2}. \quad (83)$$

Using equations (76) and (75), the phase shift of the surface topography with regard to the basal perturbations is found to be

$$\tan \phi_{SB} = \frac{-t_d^{-1} + e^{-t/t_d} (t_d^{-1} \cos(t/t_p) - t_p^{-1} \sin(t/t_p))}{t_p^{-1} - e^{-t/t_d} (t_p^{-1} \cos(t/t_p) + t_d^{-1} \sin(t/t_p))}. \quad (84)$$

with $\phi_{SC} = \pi + \phi_{SB}$.

[61] The steady state limit of expressions (82) and (84) has previously been derived by Jóhannesson [1992]. Reeh [1987] solved the steady state no-slip case. Both authors use a different definition of the Fourier transform from the one used here so the sign of their expression differs from the $t \rightarrow +\infty$ limit of equation (84).

5.2.1. Comparison of Amplitude Ratios and Phase Shifts for Long and Arbitrary Waves

[62] Equations (82) and (84) should be compared to equations (26) and (28) which give the amplitude ratio and the phase shift for long wavelengths. Expression (84) is formally identical to the corresponding expression for long waves (equation (28)), but with t_d and t_p in place of $\bar{t}_d$ and $\bar{t}_p$, respectively.

[63] Figure 3 shows the amplitude ratio and the phase shift for both long waves (thin lines) and waves of arbitrary wavelengths (thick lines) for a B perturbation. The most significant difference between the amplitude ratios for waves having long and arbitrary wavelengths, is that for long waves there is no local maximum in the amplitude ratio as a function of wavelengths. Equation (82) predicts such a local amplitude transfer maximum for a slip ratio higher than some minimum value. In Figure 3, for $C^{(0)} = 10$, such a maximum can be seen. For high sliding ratios ($C^{(0)} > 10$), the maximum of the amplitude ratio is at wavelengths around three to 10 times the mean ice thickness. For $C^{(0)} = 10$ and $\alpha = 3^\circ$ about 70% of the amplitude of a sinusoidal bedrock undulation is transferred to the surface.

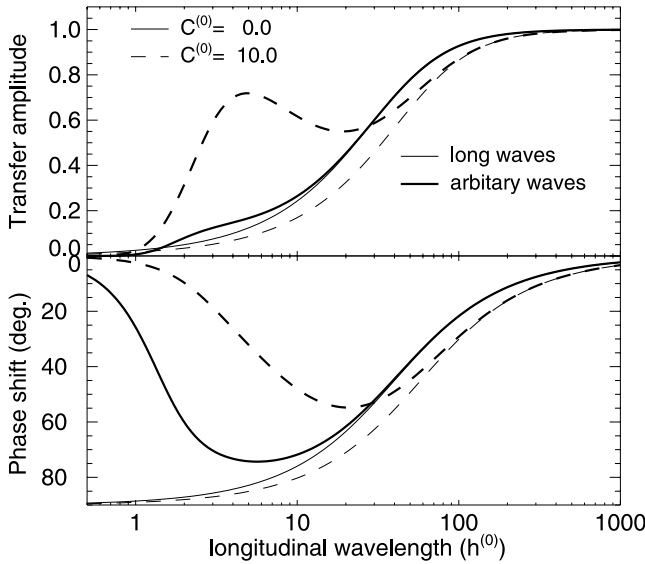


Figure 3. Steady state amplitude ratios $|T_{ZsZb}|$ and phase shifts for a perturbation in basal topography for a mean surface slope of 3° and two different slip ratios. The thin curves follow from equations (26) and (28) that were derived for long waves. The more accurate thick curves were plotted using equations (82) and (84) and are valid for waves of arbitrary wavelengths.

For $C^{(0)} = 0$, the two amplitude ratio curves in Figure 3 are rather similar, except for λ_x less than about ten. For $C^{(0)} = 10$ there are, on the other hand, considerable differences in predicted amplitude ratio for λ_x less than 40.

[64] The amplitude ratio and phase shift between surface and a basal slipperiness perturbation for long and arbitrary wavelength waves are shown in Figure 4. Neither long nor arbitrary wavelength waves exhibit a local maximum in amplitude ratios. There is a monotonic increase in amplitude ratios with increasing longitudinal wavelength toward the maximum of $C^{(0)}/(2(1 + C^{(0)}))$. Again the difference in predicted amplitude ratios and phase shifts for long and arbitrary wavelength waves increases with increasing slip ratios.

5.2.2. Phase Shifts

[65] In the steady state limit and for $C^{(0)} \rightarrow +\infty$, the phase shift ϕ_{SB} is equal to zero. Thus, if all the forward motion is due to basal sliding, the surface undulations will eventually be in phase with the basal perturbations. Another interesting limit is the high frequency ($k_x \rightarrow +\infty$) steady state limit for which $\phi_{SB} = 0$. At very short wavelengths ($\lambda_x \ll 1$) the surface disturbances are therefore in phase with the basal perturbations. Here the long-wave theory incorrectly predicts a phase shift of $-\pi/2$ (Figure 3 and equation (28)).

[66] From equations (55) and (57) it follows that v_x and v_z are always $\pi/2$ out of phase with respect to each other. The basal boundary conditions (43) and (44) imply that for symmetrical basal perturbations, v_x is symmetrical and v_z antisymmetrical. Since for $t = 0$ all fields are either symmetrical or antisymmetrical throughout the depth, v_x and v_z must also be symmetrical and antisymmetrical, respectively, along the surface. This requires a surface amplitude phase shift of $\pm\pi/2$ depending on the overall direction of flow.

From equation (84) it can be seen directly that at $t = 0$ the phase shift ϕ_{SB} is indeed equal to $-\pi/2$.

[67] Some previous authors [Hutter, 1983; Balise, 1987] considered the initial response of the surface to bedrock undulations and, correctly, predicted a phase shift of $-\pi/2$. For $t > 0$ the terms $v_x \partial_x s + v_y \partial_y s$ in the kinematic boundary condition at the surface, however, destroy the symmetry of the problem with respect to the transformation $v_i \mapsto -v_i$ and $h \mapsto -h$, and the surface phase shift is no longer necessarily equal to $-\pi/2$ for $t > 0$.

5.3. Surface Velocities

[68] The amplitude ratios of longitudinal (V_x) and transverse (V_y) velocities for a B perturbation are shown in Figures 5a and 5b. The amplitude ratio of V_x increases as a function of longitudinal wavelength ($\theta = 0$) from zero toward a maximum value, and then decreases toward zero again (Figure 5a). For $\lambda_x \rightarrow 0$, the anomalous flow is concentrated at the bed and does not reach up to the surface. Because both slope and thickness variations tend to zero as $\lambda_x \rightarrow \infty$, as discussed above, there are also no perturbations in surface velocities in this limit.

[69] For transverse waves (Figure 5b), where $\theta = \pm\pi/2$, the situation for $\lambda \rightarrow \infty$ is different as the ice then flows in x direction along longitudinal ridges and troughs. In this case, the phase shift is equal to $-\pi$ with highest velocities where the ice thickness is largest. For any given wavelength, there is an orientation (θ) of the sinusoidal B perturbation for which the anomalous flow is strictly transverse. The transverse velocities are, generally speaking, largest for $\theta \approx \pm\pi/4$ (Figure 5b).

[70] For a $C^{(0)}$ basal perturbation the flow is never strictly transversal (Figure 5c) and, as expected, the longitudinal velocities do not approach zero as λ_x goes to infinity (Figure 5c, $\theta = 0$). The largest transverse velocities (Figures 5b and 5d) are reached at rather large wavelengths of about 40–100 ice thicknesses, for these particular slope and slip ratio values.

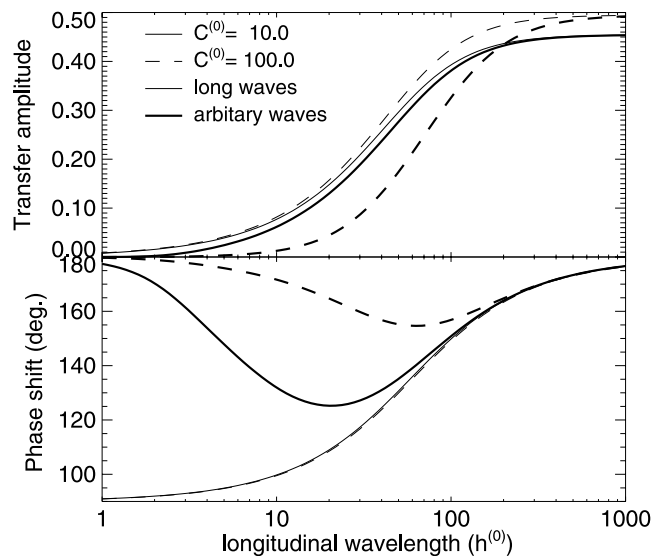


Figure 4. Same as in Figure 3 but for a basal sliding perturbation instead of a bedrock perturbation. The curves follow from equations (27), (28), (83), and (84).

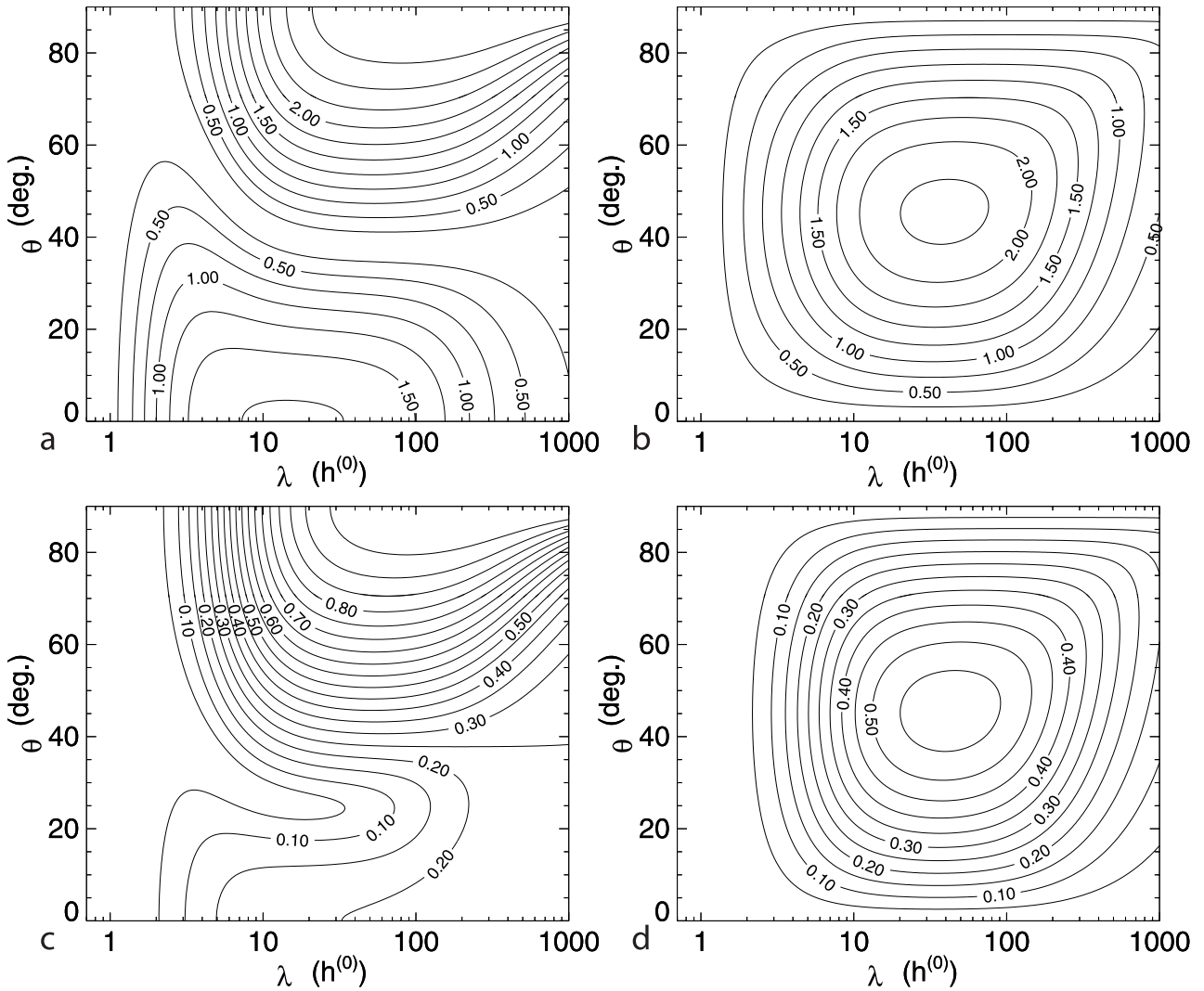


Figure 5. Steady state amplitude ratios of (a and c) longitudinal and (b and d) transverse surface velocities for both B (Figures 5a and 5b) and C basal (Figures 5c and 5d) perturbations as a function of wavelength λ and angle θ between the wave number vector the crests of the bedrock perturbation. The surface slope is 1° and $C^{(0)} = 1$.

[71] All amplitude ratios for both B and C perturbations vary smoothly as $\lambda \rightarrow \infty$ for any value of θ . There is no physical or mathematical difficulty encountered in this limit for three dimensional flow as suggested by Jóhannesson [1992]. The discontinuity in the amplitude ratio $|T_{SB}|$ as a function of the wave numbers at $k_x = k_y = 0$, discovered for the first time by him, and the fact that

$$\lim_{k_x \rightarrow 0} \lim_{k_y \rightarrow 0} |T_{SB}| \neq \lim_{k_y \rightarrow 0} \lim_{k_x \rightarrow 0} |T_{SB}|$$

is of no real concern as these two limits correspond to two different physical situations, that is $\lambda \rightarrow \infty$ for $\theta = 0$ and $\theta = \pi/2$, respectively.

5.4. Transient Behavior

[72] The transient behavior is determined by the two real valued functions t_p and t_d . The function t_d is the e -folding time of all amplitude perturbations and determines how quickly steady state is reached. The rate of phase shift is

determined by t_p . For S perturbations t_d is the decay time and $2\pi t_p$ the time period of one oscillation.

[73] These timescales are fixed by the positions of the poles in the complex plane. The denominators of the transfer functions are independent of the nature of the basal perturbations, as mentioned in section 4.9, and the associated timescales are therefore a property of the medium itself. The response of the medium to C and B perturbations is of course not identical, because the amplitudes of the anomalous fields and their relative phase shifts with respect to the applied basal perturbations differ in a number of important aspects as discussed below.

[74] In the limit of long wavelengths, the solutions of the linearized kinematic wave equation given in section 3.2 are recaptured. The timescales t_p and t_d , for example, become equal to $\bar{t}_p$ and $\bar{t}_d$ given by equations (22) and (23), and the transfer functions (75) and (76) also approach the corresponding solutions for long waves (first and second term on the right-hand side of equation (18)).

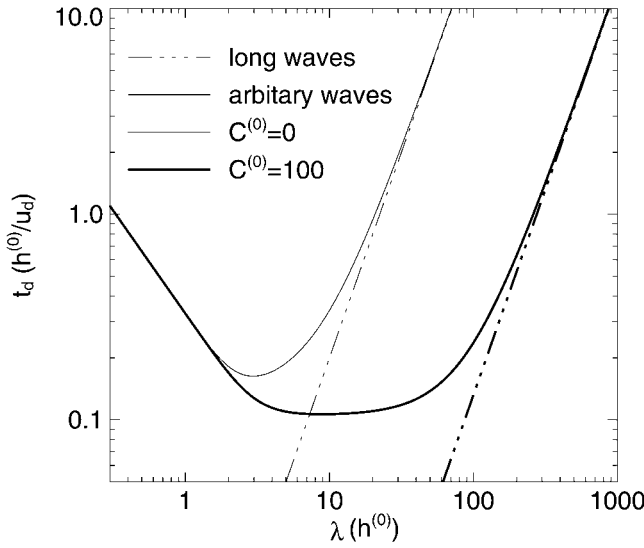


Figure 6. Decay timescale t_d for sinusoidal surface waves as a function of wavelength λ for a surface slope of 3° . The solid lines are based on equation (72) valid for waves of arbitrary wavelengths, and the dash-dotted lines are based on equation (22) for long waves. The surface slope is 3° . The decay time is not dependent on the orientation (θ) of the wave crests with respect to the mean flow direction.

[75] The decay timescale t_d is shown in Figure 6 for slip ratios of zero and ten, and for a surface slope of 3° (solid lines). The timescales are given in the units of mean thickness divided by mean deformational velocity. For typical alpine glaciers one such time unit corresponds to some 5 to 10 years. A convenient way of seeing if the predictions of long-wavelength and the arbitrary wavelength theories are similar for a given wavelength, is to compare t_d and $\bar{t}_d$ as given by equations (72) and (22).

[76] The long-wave decay time $\bar{t}_d$ decreases monotonically with decreasing wavelength. The dependency of t_d on λ is much more complex. At short wavelengths t_d decreases with increasing wavelength, and for moderate to high slip ratio an intermediate range of wavelengths can be identified over which t_d is independent of λ . As λ is increased further, t_d eventually increases again and approaches the long-wavelength limit.

[77] In the absence of basal sliding, the differences between t_d and $\bar{t}_d$ are limited to rather short wavelengths of less than about 10 ice thicknesses, which is in line with what one would intuitively expect. This is, however, not true for high slip ratios as can be seen from Figure 6. For the long-wavelength theory to give an accurate description of glacier flow it is, thus, not sufficient that the wavelengths are long. The slip ratio may also not be too large. This can be understood to be a consequence horizontal stresses being transmitted over long distances at high slip ratio. This enhancement in stress transmission with increasing sliding velocities has been discussed by various authors [e.g., Lliboutry and Reynaud, 1981; Kamb and Echelmeyer, 1986] but the role of the slip ratio as the pertinent variable has not been stressed sufficiently. Schoof [2003] analyzed the flow of ice streams theoretically by splitting the flow field up into an outer and inner flow field. The advantage

of this approach is that the mean basal velocity becomes a quantity that is determined from an appropriate matching between the inner and the outer problems, whereas here the local mean basal drag is determined from the long-wavelength limit and is independent of the actual local bed properties. The inner problem considered by Schoof [2003] is a special case of the arbitrary wave problem consider here, as can be seen by comparing the form of the linearized boundary conditions. Lateral stress transmission has been discussed by Raymond [1996] in his theoretical analysis of ice stream margins.

5.5. Propagation and Decay of Surface Waves

[78] The transient evolution of the surface geometry is given by the inverse Fourier transform of the product of T_{SS} (equation (78)) and the surface profile at $t = 0$, that is

$$S(t, \mathbf{r}) = \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} d\mathbf{k} S(t=0, \mathbf{k}) \cdot e^{-i(\mathbf{k} \cdot \mathbf{r} - w(\mathbf{k})t)} \quad (85)$$

where $\mathbf{k} = (k_x, k_y)$, $\mathbf{r} = (x, y)$.

[79] The dispersion relation

$$w(\mathbf{k}) = t_p^{-1}. \quad (86)$$

has two branches corresponding to positive and negative frequencies. The group velocity is

$$\mathbf{V}^g = (\partial_{k_x} w, \partial_{k_y} w) \quad (87)$$

which for transverse waves gives

$$\frac{V_x^g}{V_s^{(0)}} - 1 = \frac{(1 - 2k_x^2 C^{(0)})C_h^2 - k_x(k_x + 2C_h S_h)}{(C_h^2 + k_x^2 V_s^{(0)})^2 + k_x C^{(0)} C_h S_h \tilde{w}}, \quad (88)$$

with

$$\tilde{w} := C_h k_x C^{(0)} S_h + 2C_h^2 + 2k_x^2 V_s^{(0)} \quad (89)$$

where $C_h := \cosh k_x$ and $S_h := \sinh k_x$. For long transverse waves $w = 2V_s^{(0)}k_x$, and for short waves $w = V_s^{(0)}k_x$, where $V_s^{(0)} = 1 + C^{(0)}$ is the mean surface velocity. Both short and long waves are, therefore, nondispersive with phase and group velocities given by $V_s^{(0)}$ and $2V_s^{(0)}$, respectively. In general, however, both the phase and the group velocities depend on the wavelength (Figure 7). The phase velocity points in the same direction as the wave number vector $\mathbf{k}$ and the phase speed is (V^p) given by

$$V^p = k^{-1} w(k \cos \theta, k \sin \theta) \quad (90)$$

At short wavelengths and for $\theta = 0$ the phase speed is equal to the mean surface velocity. Such small sized surface perturbations are therefore passively advected as pointed out by Lick [1970], Thompson [1979], Landon and Raymond [1978], and Jóhannesson [1992]. With increasing transverse wavelength the phase speed increases toward $2(1 + C^{(0)})$. This same long-wavelength limit can also be derived directly from equation (15) and is a well-known result [Nye, 1958; Nye, 1960; Weertman, 1958]. Landon and Raymond [1978]

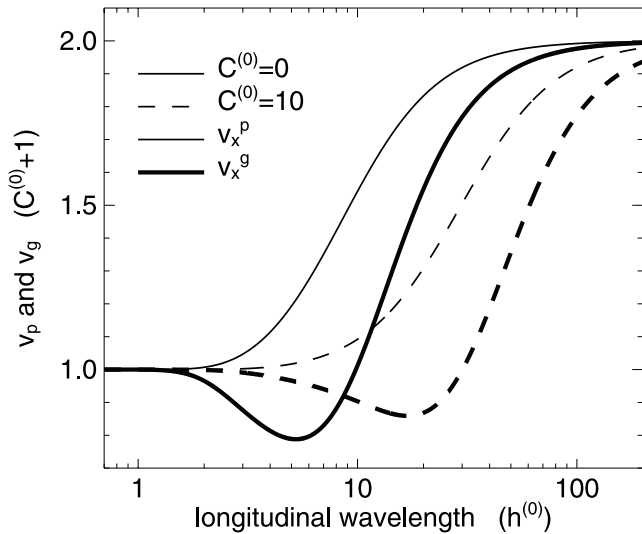


Figure 7. Group and phase velocities (v_x^g , v_x^p) for $\theta = 0$ as a function of λ for two different values of the slip ratio $C^{(0)}$.

determined theoretically the decay time and phase velocity as function of transverse wavelength for $C^{(0)} = 0$.

[80] For $\theta = 0$ the group velocity is always smaller than the phase velocity except for short and long (transverse) wavelengths where these two velocities are equal. For a range of transverse wavelengths the group velocity is even smaller than the average surface velocity as can be seen in Figure 7.

[81] Whereas the phase velocity is always directed normal to the crests of any sinusoidal surface undulations, that is parallel to the wave number vector $\mathbf{k}$, this is not the case for the group velocity. Figure 8 shows how the two components of the group velocity (V_x^g and V_y^g) change as the orientation of a sinusoidal surface wave with the respect to the x axis (θ) is varied. The phase speed V^p is also shown, as is the angle ψ between the group velocity and the main flow direction

$$\psi = \arctan V_y^g / V_x^g.$$

The phase speed V^p depends on θ as $\cos \theta$, with V^p equal to zero for $\theta = \pm\pi/2$ as it should be. In general, the x component of the group velocity (V_x^g) increases with θ although this increase goes toward zero for both short and long waves. The group velocity for longitudinal surface ridges is, thus, somewhat larger than for corresponding transverse ridges. The transverse component of the group velocity (V_y^g) is negative for positive θ reaching a maximum magnitude for $\theta = \pm\pi/4$. The direction of the group velocity never deviates by more than a few degrees from the main flow direction, and is aligned exactly with the direction of mean flow at both short and long wavelengths.

[82] As emphasized by both Lick [1970] and Jóhannesson [1992] transverse kinematic surface waves on glaciers are strongly diffusive. The e -folding time of surface waves is t_d (equation (78)) and the number of periods (N) it takes the amplitude to be reduced by the factor e is

$$N = \frac{1}{2\pi} \frac{t_d}{t_p}. \quad (91)$$

This number is, for realistic values of slopes and slip ratios for glaciers and ice sheets, usually much smaller than unity [Jóhannesson, 1992]. It can be shown using equation (91) together with equations (72) and (73) that for $|\theta| > 0$ the attenuation is even greater than for $\theta = 0$. The smallness of N shows that viscous damping of waves on glaciers is very strong and Lick [1970] used the term ‘diffusive kinematic wave’ to emphasize the strongly diffusive character of such waves.

[83] As an illustration of the transient evolution of surface disturbances on glaciers Figure 9 depicts the evolution of a two-dimensional Gaussian surface bump. At $t = 0$ the surface perturbations is a Gaussian function having standard deviations of $\sigma_x = 0.5$ and $\sigma_y = 0.5$ in x and y directions, respectively. This wave is dispersive with the long wavelength components traveling at a phase speed of $2V_s^{(0)} = 2(1 + C^{(0)}) = 4$. The short wavelength components can be seen to be advected with the ice with a velocity equal to $V_s^{(0)}$ and to decay rather quickly forming a short-lived “wiggle” on the top of a much broader general surface profile. After the wiggle has disappeared the maximum of the surface profile travels downstream with a velocity equal to the long-wavelength group velocity. Although a kinematic wave can be seen in Figure 9, it is nevertheless presumably misleading to speak of the redistribution of the mass on glaciers and ice sheets in terms of a wave propagation. What Figure 9 shows can be more accurately be described as a diffusion process.

[84] Only for very steep glaciers having surface slopes larger than about 10° and large sliding ratios (>500) does N become larger than five for wavelengths less than about

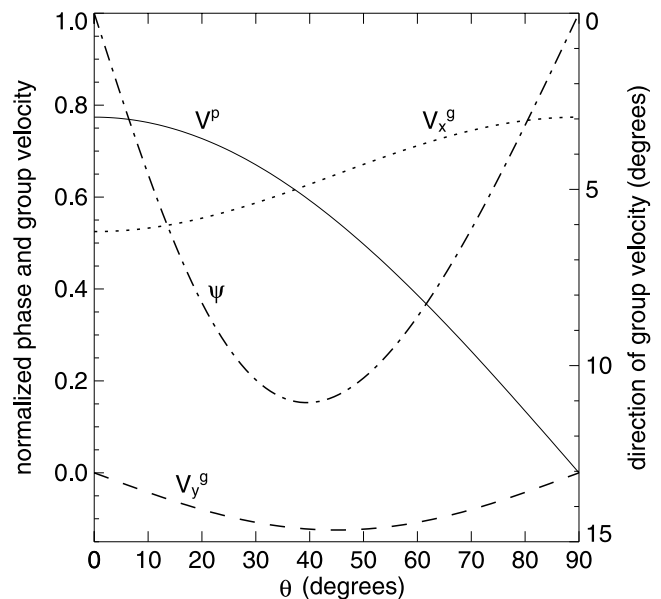


Figure 8. Phase speed (V^p), both components of group velocity (V_x^g , V_y^g), and the direction of the group velocity (ψ) with respect to the x axis as a function of the angle θ between the x axis and the wave number vector. The phase speed and the group velocity are normalized with their long-wavelength limits for $\theta = 0$, which is $2(C^{(0)} + 1)$. The wavelength of the sinusoidal surface wave is $\lambda = 50$ and the slip ratio is $C^{(0)} = 50$.

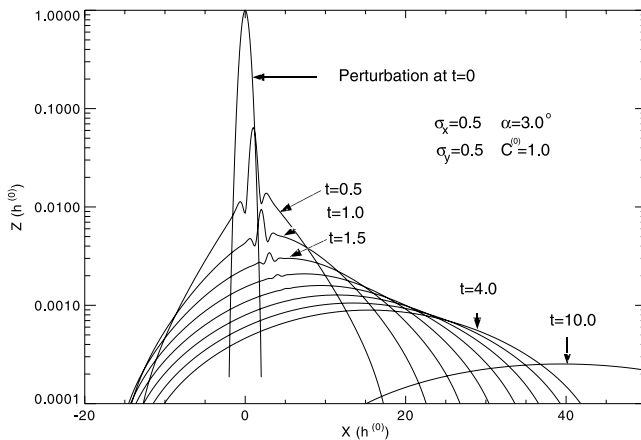


Figure 9. Transient evolution of a Gaussian surface perturbation. The center of the perturbation is at $(X, Y) = (0, 0)$ at $t = 0$. The mean flow velocity is from left to right and the mean surface velocity is $C^{(0)} + 1 = 2$. Longitudinal profiles along the medial line ($Y = 0$) are shown for a number of different time steps.

10 $h^{(0)}$. This follows from equation (91) using equations (72) and (73) for t_d and t_p . For most known glaciers it will, thus, be essentially impossible to observe kinematic waves. Numerical calculations of the transient evolution of finite amplitude surface undulations for both linear and power law medium show that this remains true even when nonlinear effects are fully taken into account [Lambert, 2001]. In spite of this, the glaciological literature contains detailed references to observations of such waves [Lliboutry and Reynaud, 1981; Sturm *et al.*, 1986] and there is no reason to doubt that such waves do indeed exist. Theoretically such traveling waves can not be caused by surface processes only. They must be related to some other mechanism, such as transient basal processes, a high sensitivity of basal velocities to small changes in basal stress distribution, or till wave propagation as suggested by Hindmarsh [1998b]. From a comparisons of changes in geometry and surface velocity Sturm *et al.* [1986] concluded that a surface wave observed on Mount Redoubt, Alaska, was affecting basal sliding velocities. Lliboutry and Reynaud [1981] also stressed the effects of basal processes on the propagation of surface waves.

[85] It is interesting to note that if equations (22) and (23) for the long-wavelength timescales $\bar{t}_d$ and $\bar{t}_p$ are used instead of t_d and t_p in equation (91), N turns out to be, even for steep slopes, smaller than unity. The linearized long-wavelength equation for linear medium (equation (15)) and the corresponding nonlinear version [Nye, 1960] does clearly not predict that surface waves propagate any significant distances downstream despite various such claims found in the literature.

[86] The dispersive property of small-amplitude surface waves makes it unlikely that nonlinear effects will accumulate and give rise to strongly nonlinear effects such as shock waves. Lambert [2001] used a commercial numerical finite element model which was tailored for glaciological applications by Leysinger [1998] and Gudmundsson [1999] to calculate transient evolution of long waves for various

slopes. Only for unrealistic mean surface slopes higher than about 45° where shock waves observed to develop.

5.6. Transient Surface Response to Basal Perturbations

[87] Figure 10 shows the transient response of the surface to a Gaussian bedrock perturbation. Initially, the surface response is antisymmetrical about $x = 0$, as it must be for Stokes flow. Upstream from the basal perturbation the ice piles up and a peak is formed at the surface, while on the downstream side a surface depression is formed. With time the amplitude of the upstream peak increases. The depth of the depression initially increases with time too, but then decreases as it gets advected downstream. Eventually, the peak is located almost exactly above the top of the bedrock perturbation, while the depression has disappeared at infinity, leaving a shallow surface 'stripe' stretching out downstream over considerable distances.

[88] As seen from Figures 3 and 4, the basal-to-surface transfer depends strongly on the value of the slip ratio $C^{(0)}$. In general, the surface response becomes larger and more asymmetrical with increasing slip ratio. In particular, the depth of the stripe seen in Figure 10a decreases ever more slowly in down-flow direction as the value of $C^{(0)}$ increases.

[89] From equations (82) and (83) one sees that a sudden onset of a sinusoidal basal perturbation can, for certain ratios of t_d to t_p , give rise to amplitude oscillations at the surface. This type of transient response can be referred to as kinematic oscillations in analogy to the term kinematic wave. Example for a kinematic oscillation is given in Figure 11a. The timing of the first amplitude maximum is related to t_p . In Figure 11b the wavelength is somewhat larger and the t_p/t_d ratio smaller than in Figure 11a, and as a result there is a monotonic increase in the surface amplitude over time, which is essentially determined by the timescale t_d .

6. Conclusions

[90] The slip ratio determines the horizontal spatial scale of transmission of stress gradients, and affects the flow mechanics in a number of important ways. At small to moderate slip ratios (zero to ten), the effects of a local basal perturbation on the surface are limited to a fairly small region (<10), and the relation between surface and small-amplitude basal disturbances adequately described by the long-wavelength approximation. At high slip ratios (>100) the long-wavelength approximation is poor, even at wavelengths that are 50 times the mean ice thickness. For example, for slip ratios higher than about one hundredth and for mean surface slopes of 3° , the amplitude ratios calculated with these two theories, differ by about a factor of two and the decay timescales by an order of magnitude.

[91] Above a certain minimum slip ratio, the amplitude ratio between bedrock and surface geometry reaches a local maximum as a function of transverse wavelength. In general, the transfer amplitudes increase with increasing slip ratio.

[92] For fast flowing ice streams, where most of the forward motion is due to basal sliding, one can expect the flow to be strongly nonlocalized, and stress gradients to be transmitted over large distances as compared to the mean ice thickness. It is, thus, not surprising that ice stream margins can have a decisive effect on the flow mechanism of an

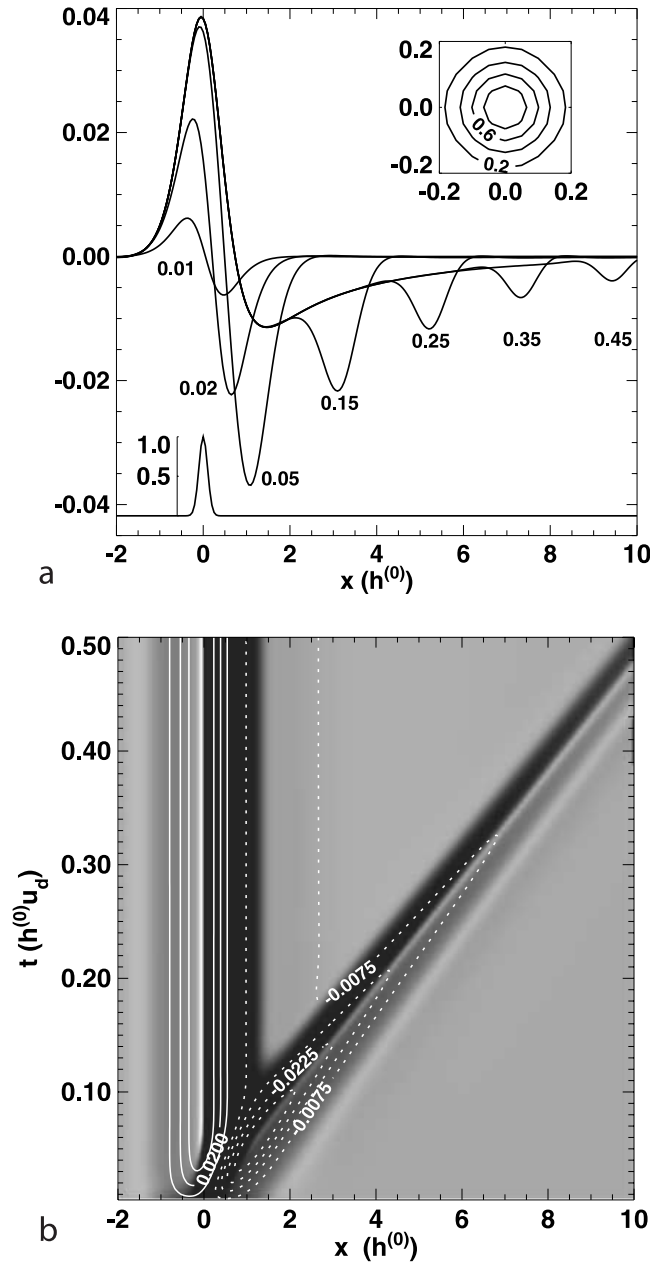


Figure 10. Transient response of the surface to a Gaussian-shaped bedrock undulation for $C^{(0)} = 20$ and $\alpha = 3^\circ$. (a) set of curves showing the surface shape along a longitudinal profile running across the medial line ($y = 0$) for a number of time steps, starting with $t = 0.01$ and ending with $t = 0.45$. The mean flow direction is from left to right. For reference, a corresponding longitudinal profile along the bed is given by the curve in the lower part of the figure. The contour lines of the Gaussian bedrock undulation is shown in map plane view in the inset. Note the different vertical scales of the surface disturbances (left ordinate) and the bedrock undulation (inset axis). (b) Space-time diagram of the time evolution of the surface disturbances.

active ice stream despite typical thickness-to-width ratios as small as 0.01. For ice streams the controlling length scale can be expected to the width rather than the thickness. One must also expect basal variability to be transmitted quite

easily toward the surface, especially for transverse wavelengths within the range of about 5 to 20 ice thicknesses.

[93] Slipperiness and bedrock perturbations affect the surface geometry and surface velocities differently. Information about two independent fields, for example surface geometry and one component of surface velocities, is needed in order to distinguish between these two different types of basal boundary conditions from surface data. Because of the increase in amplitude ratio with increasing slip ratio, regions having high slip ratios are best suited for performing such an inversion.

[94] The transient evolution is governed by the decay and propagation timescales (t_d and t_p). Decay and propagation are properties of the medium and not affected by the nature of the external perturbation. Additional timescales can be expected to be introduced if other boundaries apart from the surface are allowed to evolve, as is, for example the case, in the presence of basal cavitation.

[95] Small amplitude surface waves on glaciers are both strongly diffusive and strongly dispersive. The above linear analysis clearly can not describe any nonlinear features of

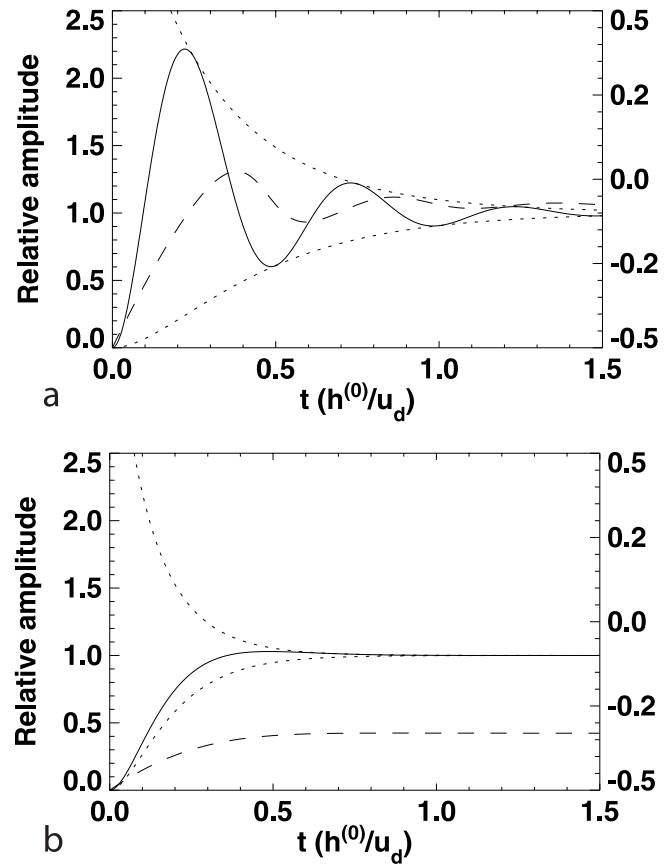


Figure 11. Transient response of the surface to a sinusoidal basal bedrock and slipperiness perturbation for (a) $\lambda_x = 1$ and (b) $\lambda_x = 3$. Other parameters are $\alpha = 3^\circ$, $C^{(0)} = 1$ and $\lambda_y = \infty$. The solid lines give the relative amplitudes of the sinusoidal surface undulations as functions of time (terms in braces in equations (82) and (83)). The phase shifts (in units of π) are shown as dashed lines. The two dotted curves in both Figures 11a and 11b are given by the functions e^{-t/t_d} ($e^{-t/t_d} \pm 2$). They give the maximum upper and lower limits of the amplitude response.

surface waves. However, the strongly dispersive nature of small amplitude waves makes it unlikely that nonlinear effects can cumulate sufficiently to give rise to shock waves and this is supported by numerical calculations.

[96] Mass redistribution on glaciers and ice sheets, following a small-amplitude perturbation in surface geometry, is primarily a decay process, and it is misleading, although not mathematically incorrect, to describe it in terms of kinematic wave propagation. The decay timescale is considerably shorter than the time it takes a kinematic wave to travel down a typical valley glacier. For this reason, one does not expect the response time of glaciers to scale directly with the ratio of surface velocities to the glacier length.

[97] For realistic slopes and most transverse wavelengths, the number of periods required for the amplitude of a surface wave to be reduced by the factor e is smaller than unity. One must therefore consider the possibility that observations of surface waves on glaciers are indication of transient basal processes rather than kinematic (surface) waves.

Appendix A: Derivation of the First-Order Sliding Boundary Condition

[98] The derivation of the first-order boundary conditions is illustrated using the sliding condition of the B perturbation problem as an example. Index notation is used throughout. Primed components are with respect to a rotated coordinate system $\hat{\mathbf{e}}'_1\hat{\mathbf{e}}'_2\hat{\mathbf{e}}'_3$ with $\hat{\mathbf{e}}'_1$ and $\hat{\mathbf{e}}'_2$ spanning the tangential plain at some point of the undulating bed. The vectors $\hat{\mathbf{e}}'_i$ for $i = 1, 2, 3$ are unit vectors with $\hat{\mathbf{e}}'_3$ normal to the lower surface and $\hat{\mathbf{e}}'_i \cdot \hat{\mathbf{e}}'_j = \delta_{ij}$.

[99] The sliding law has a particularly simple form in this tilted coordinate system ($x'y'z'$) or $v'_1 = c(x', y')$ σ'_{13} and $v'_2 = c(x', y')$ σ'_{12} , where the prime denotes the components in the $x'y'z'$ system.

[100] For slope variations around the mean slope small compared to unity one finds that the components of the Cauchy stress tensor σ'_{ij} in the rotated system are related to σ_{ij} through $\sigma'_{ij} = R_{ik}R_{jl}\sigma_{kl}$, where

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & \partial_x b \\ 0 & 1 & \partial_y b \\ -\partial_x b & -\partial_y b & 1 \end{pmatrix} \quad (\text{A1})$$

is a matrix describing the infinitesimal transformation.

[101] The sliding condition then reads $V'_1 = c(x', y')$ Σ'_{13} , and $V'_2 = c(x', y')$ Σ'_{12} , where $\Sigma'_{ij} = R_{ik}R_{jl}\Sigma_{kl}$, and $V'_i = R_{ij}V_j$. The Taylor expansions

$$\Sigma_{ij} = \Sigma_{ij}^{(0)} + \varepsilon F_b(X, Y) \partial_Z \Sigma_{ij}^{(0)} + \varepsilon \Sigma_{ij}^{(\varepsilon)} + O(\varepsilon^2), \quad (\text{A2})$$

and

$$V_i = V_i^{(0)} + \varepsilon F_b(X, Y) \partial_Z V_i^{(0)} + \varepsilon V_i^{(\varepsilon)} + O(\varepsilon^2), \quad (\text{A3})$$

are used to transform the boundary conditions along the undulating boundary to the boundary of the zeroth-order

solution, where $B(X, Y) = B^{(0)} + \varepsilon F_b(X, Y)$. In equations (A2) and (A3) the left-hand sides are evaluated at the actual undulating boundary and the right-hand sides at the nonundulating boundary of the zeroth-order solution.

[102] Using equation (A1) to calculate, to first order in ε , the projection of the stress vector $\mathbf{S}(\hat{\mathbf{e}}'_3)$ along $\hat{\mathbf{e}}'_1$ leads to

$$\Sigma'_{13} = \Sigma_{13} + \varepsilon(\Sigma_{33} - \Sigma_{11}) \partial_X F_b - \Sigma_{12} \partial_Y F_b + O(\varepsilon^2), \quad (\text{A4})$$

and for the projection of $\mathbf{S}(\hat{\mathbf{e}}'_3)$ along $\hat{\mathbf{e}}'_2$ to

$$\Sigma'_{23} = \Sigma_{23} - \varepsilon \Sigma_{21} \partial_X F_b + O(\varepsilon^2). \quad (\text{A5})$$

The relations between the components of the velocity vector in the primed and the unprimed systems are

$$V'_1 = V_1 + \varepsilon V_3 \partial_X F_b + O(\varepsilon^2), \quad (\text{A6})$$

and

$$V'_2 = V_2 + \varepsilon V_3 \partial_Y F_b + O(\varepsilon^2), \quad (\text{A7})$$

Inserting equations (A4) and (A6) into the sliding law and using the Taylor expansions (A2) and (A3) leads to the first-order equation

$$\begin{aligned} V_1^{(\varepsilon)} + F_b \partial_Z V_1^{(0)} + V_3^{(0)} \partial_X F_b \\ = C^{(0)} \left\{ \Sigma_{13}^{(\varepsilon)} + F_b \partial_Z \Sigma_{13}^{(0)} + (\Sigma_{33}^{(0)} - \Sigma_{11}^{(0)}) \partial_X F_b - \Sigma_{12}^{(0)} \partial_Y F_b \right\} \end{aligned} \quad (\text{A8})$$

It then follows from the properties of the zeroth-order solution ((34) to (38)) that

$$(2 + C^{(0)}) F_b + V_1^{(\varepsilon)} = C^{(0)} \Sigma_{13}^{(\varepsilon)}. \quad (\text{A9})$$

From (A5) and (A7), the Taylor expansions (A2) and (A3), the sliding law, and the properties of the zeroth-order solution it furthermore follows that

$$V_2^{(\varepsilon)} = C^{(0)} \Sigma_{23}^{(\varepsilon)}. \quad (\text{A10})$$

Equation (A9) represents the “ ε part” of equation (44) which was stated without derivation, and equation (A10) the ε part of equation (45).

Appendix B: Expressions Entering the S Transfer Function

[103] The following constants appearing in the text are defined as

$$\begin{aligned} \tilde{g} := & \left\{ \left[(C^{(0)} k^2 + 3k) C^{(0)} + 2 \right] k^3 [\cosh(k)]^2 \right. \\ & + \left[(2 + k C^{(0)}) (1 + C^{(0)}) - C^{(0)2} \right] k^5 \tilde{h} \\ & \left. - 3 C^{(0)} k^4 [\cosh(k)]^2 - C^{(0)} (1 + C^{(0)}) k^6 \right\} \end{aligned}$$

$$\tilde{h} := [\sinh(k) + \cosh(k)] \cosh(k)$$

$$\begin{aligned} \tilde{o} := & -k_x \left(C^{(0)}k + 2 \right) \left(C^{(0)}k + 1 \right) k_y e^{6k} \\ & + \left\{ 8k_y \left(1 + C^{(0)} \right) k^3 \cot \alpha - k_x k_y k \right. \\ & \cdot \left[-8 + 8C^{(0)}k^2 \left(1 + C^{(0)} \right) - 3C^{(0)} \right] \left\{ e^{4k} + e^{2k} \right\} \\ & + \left\{ 4C^{(0)}k k_y \left(1 + C^{(0)} \right) k^3 \cot \alpha - k_x k_y \right. \\ & \cdot \left[k^2 \left(C^{(0)} \left(12 + C^{(0)} \right) + 16 \right) + 2 \right] \left\{ e^{4k} - e^{2k} \right\} \\ & \left. + k_x \left(C^{(0)}k - 1 \right) \left(C^{(0)}k - 2 \right) k_y \right\} \end{aligned}$$

$$\begin{aligned} \tilde{p} := & \left(C^{(0)}k + 2 \right) \left(C^{(0)}k + 1 \right) k^3 e^{6k} \\ & + \left[3 + 4k^2 \left(1 + C^{(0)} \right) \right] C^{(0)}k^4 \left(e^{4k} - e^{2k} \right) \\ & + \left[6 + 8k^2 \left(1 + C^{(0)} \right) - C^{(0)2}k^2 \right] k^3 \left(e^{4k} + e^{2k} \right) \\ & + \left(C^{(0)}k - 1 \right) \left(C^{(0)}k - 2 \right) k^3 \end{aligned}$$

$$\begin{aligned} \tilde{q} := & \left[\left(kC^{(0)} + 2 \right) \tilde{h} - kC^{(0)} \right] k^3 k_x \left(1 + C^{(0)} \right) \cot \alpha \\ & + \left(kC^{(0)} + 2 \right) \left(kC^{(0)} + 1 \right) \left(k^2 + k_y^2 \right) \tilde{h} [\cosh(k)]^2 \\ & + \left\{ 2k^3 k_y^2 C^{(0)2} + k \left[(-3k - 2k^2 + 3) k_x^2 \right. \right. \\ & \left. \left. + 2k^2 (k^2 - 2 + 2k) \right] \cdot C^{(0)} \right\} \\ & + 2k \left[(1 - 2k) k_x^2 + 2k^3 \right] \tilde{h} + \left(C^{(0)2}k^2 + 2 \right) \\ & \cdot \left(k_x^2 - 2k^2 \right) [\cosh(k)]^2 - k^2 \left[4k^2 \left(1 + C^{(0)} \right) \right. \\ & \left. - 3k_x^2 C^{(0)} - 4k_x^2 \right] \end{aligned}$$

Notation

α	surface slope.
α_x, α_y	surface slopes in x and y directions, respectively.
β	slipperiness amplitude parameter.
δ	thinness parameter, d/l .
ε	bedrock amplitude parameter.
η	viscosity.
θ	angle between the wave number vector and the x axis.
λ	wavelength.
λ_x	longitudinal wavelength.
λ_y	transverse wavelength.
μ	defined through $\mu := \delta \cot \alpha$.
Σ_{ij}	components of the nondimensional Cauchy stress tensor.
σ_{ij}	components of the Cauchy stress tensor.
τ_b	basal shear stress.
ϕ	phase shift.
ψ	Angle between the group velocity and the main flow direction.

B	nondimensional vertical position of bedrock.
b	vertical position of bedrock.
c	dimensional slipperiness.
$C^{(0)}$	mean slipperiness, for a constant viscosity equal to the slip ratio (u_b/u_d).
D	nondimensional thickness.
d	ice thickness.
f_i	components of body force.
k_i	components of the wave number vector.
l	horizontal length scale.
m_i	components of surface unit normal vector.
N	number of periods by which the amplitude of sinusoidal surface waves has reduced by a factor e .
n_i	components of basal unit normal vector.
t	time.
t_d	decay timescale.
t_p	propagation timescale.
$\bar{t}_d$	long-wavelength decay timescale.
$\bar{t}_p$	long-wavelength propagation timescale.
P	nondimensional pressure.
p	dimensional pressure.
S	nondimensional vertical position of surface.
s	vertical position of surface, also used as a Laplace transform variable.
$\mathcal{T}$	arbitrary wavelength transfer function.
$\bar{\mathcal{T}}$	long-wavelength transfer function.
u_b	basal sliding velocity.
u_d	deformational velocity, the contribution of internal ice deformation to the mean surface velocity.
$V_s^{(0)}$	mean surface velocity.
$\mathbf{V}^g$	group velocity.
V_i	nondimensional components of the velocity vector.
v_i	components of the velocity vector.
w	angular frequency.

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References

- Alley, R. B., In search of ice-stream sticky spots, *J. Glaciol.*, 39, 447–545, 1993.
- Balise, M. J., The relation between surface and basal velocity variations in glaciers, with application to the mini-surges of Variegated Glacier, Ph.D. thesis, Univ. of Wash., Seattle, 1987.
- Balise, M. J., and C. F. Raymond, Transfer of basal sliding variations to the surface of a linearly viscous glacier, *J. Glaciol.*, 31, 308–318, 1985.
- Budd, W. F., The longitudinal stress and strain-rate gradients in ice masses, *J. Glaciol.*, 9, 19–27, 1970a.
- Budd, W. F., Ice flow over bedrock perturbations, *J. Glaciol.*, 9, 29–48, 1970b.
- Budd, W. F., Stress variations with ice flow over undulations, *J. Glaciol.*, 10, 177–195, 1971.
- Collins, I. F., On the use of the equilibrium equations and flow law in relating the surface and bed topography of glaciers and ice sheets, *J. Glaciol.*, 7, 199–204, 1968.
- Finstervwalder, S., Die Theorie der Gletscherschwankungen, *Z. Gletscherkd. Glazialgeol.*, 1, 81–103, 1907.
- Fowler, A. C., A mathematical approach to the theory of glacier sliding, *J. Glaciol.*, 23, 131–141, 1979.
- Fowler, A. C., Waves on glaciers, *J. Fluid Mech.*, 120, 283–321, 1982.
- Fowler, A. C., and D. A. Larson, On the flow of polythermal glaciers. I. Model and preliminary analysis, *Proc. R. Soc. London, Ser. A*, 363, 217–242, 1978.

- Fowler, A. C., and D. A. Larson, On the flow of polythermal glaciers. II. Surface wave analysis, *Proc. R. Soc. London, Ser. A*, 370, 155–171, 1980.
- Fowler, A. F., Modelling the flow of glaciers and ice sheets, in *Continuum Mechanics and Applications in Geophysics and the Environment*, edited by B. Straughan et al., pp. 201–221, Springer-Verlag, New York, 2000.
- Gudmundsson, G. H., A three-dimensional numerical model of the confluence area of Unteraargletscher, Bernese Alps, Switzerland, *J. Glaciol.*, 45, 219–230, 1999.
- Gudmundsson, G. H., C. F. Raymond, and R. Bindshadler, The origin and longevity of flow-stripes on Antarctic ice streams, *Ann. Glaciol.*, 27, 145–152, 1998.
- Gudmundsson, G. H., T. Thorsteinsson, and C. F. Raymond, Inferring bed topography and stickiness from surface data on ice streams, *Eos Trans. AGU*, 82, Fall Meet. Suppl., Abstract IP21A-0687, 2001.
- Hindmarsh, R. C. A., The stability of a viscous till sheet coupled with ice flow, considered at wavelengths less than the ice thickness, *J. Glaciol.*, 44, 285–292, 1998a.
- Hindmarsh, R. C. A., Ice-stream surface texture, sticky spots, waves and breathers: The coupled flow of ice, till and water, *J. Glaciol.*, 44, 589–614, 1998b.
- Hutter, K., Time-dependent surface elevations of an ice slope, *J. Glaciol.*, 25, 247–266, 1980.
- Hutter, K., Dynamics of glaciers and large ice masses, *Annu. Rev. Fluid Mech.*, 14, 87–130, 1982.
- Hutter, K., *Theoretical Glaciology: Material Science of Ice and the Mechanics of Glaciers and Ice Sheets*, D. Reidel, Norwell, Mass., 1983.
- Hutter, K., Thermo-mechanically coupled ice-sheet response—Cold, polythermal, temperate, *J. Glaciol.*, 39, 65–86, 1993.
- Hutter, K., F. Legerer, and U. Spring, First-order stresses and deformations in glaciers and ice sheets, *J. Glaciol.*, 27, 227–270, 1981.
- Jóhannesson, T., Landscape of temperate ice caps, Ph.D. thesis, Univ. of Wash., Seattle, 1992.
- Kamb, B., and K. A. Echelmeyer, Stress-gradient coupling in glacier flow: I. Longitudinal averaging of the influence of ice thickness and surface slope, *J. Glaciol.*, 32, 267–284, 1986.
- Lambert, M., Ausbreitung wellenförmiger Störungen auf einem Gletscher, technical report, Versuchsanst. für Wasserbau, Hydrol. und Glaziol., ETH-Zürich, Zürich, Switzerland, 2001.
- Landon, J., and C. F. Raymond, Chislenniy raschet reaktivnoy poverkhnosti lednika na izmeneniya tolshchiny l'da Mater, *Glyatsiol. Issled. Khron. Obsuzhdeniya*, 32, 123–133, 1978. (English translation, Numerical calculation of adjustment of a glacier surface to perturbations of ice thickness, *Mater. Glyatsiol. Issled. Khron. Obsuzhdeniya*, 32, 233–239, 1978.)
- Leysinger, G. J.-M. C., Numerisches Blockgletschermodell in zwei Dimensionen, Diploma thesis, 64 pp., Versuchsanst. für Wasserbau, Hydrol. und Glaziol., ETH-Zürich, Zürich, Switzerland, 1998.
- Lick, W., The propagation of disturbances on glaciers, *J. Geophys. Res.*, 75, 2189–2197, 1970.
- Lliboutry, L., The glacier theory, *Adv. Hydrosol.*, 7, 81–167, 1971.
- Lliboutry, L., and L. Reynaud, 'Global dynamics' of temperate valley glacier, Mer de Glace, and past velocity deduced from Forbes' bands, *J. Glaciol.*, 27, 207–226, 1981.
- Morland, L. W., Influence of bed topography on steady plane ice sheet flow, in *Continuum Mechanics and Applications in Geophysics and the Environment*, edited by B. Straughan et al., pp. 276–304, Springer-Verlag, New York, 2000.
- Morland, L. W., and I. R. Johnson, Steady motion of ice sheets, *J. Glaciol.*, 25, 229–246, 1980.
- Nye, J. F., A theory of wave formation in glaciers, in *Symposium at Chamonix 1958-Physics of the Movement of Ice*, IASH Publ., 47, 139–154, 1958.
- Nye, J. F., The motion of ice sheets and glaciers, *J. Glaciol.*, 3, 493–507, 1959.
- Nye, J. F., The response of glaciers and ice-sheets to seasonal and climatic changes, *Proc. R. Soc. London, Ser. A*, 256, 559–584, 1960.
- Nye, J. F., On the theory of the advance and retreat of glaciers, *Geophys. J. R. Astron. Soc.*, 7, 431–456, 1963a.
- Nye, J. F., The response of a glacier to changes in the rate of nourishment and wastage, *Proc. R. Soc. London, Ser. A*, 275, 87–112, 1963b.
- Nye, J. F., The frequency response of glaciers, *J. Glaciol.*, 5, 567–587, 1965.
- Raymond, C. F., Shear margins in glaciers and ice sheets, *Ann. Glaciol.*, 42, 90–102, 1996.
- Reeh, N., Steady-state three-dimensional ice flow over an undulating base: First-order theory with linear ice rheology, *J. Glaciol.*, 33, 177–185, 1987.
- Schoof, C., Basal perturbations under ice streams: Form drag and surface expression, *J. Glaciol.*, 48, 407–416, 2003.
- Sturm, M., C. Benson, and P. MacKeith, Effects of the 1966–68 eruptions of Mount Redoubt on the flow of Drift Glacier, Alaska, U.S.A., *J. Glaciol.*, 32, 355–362, 1986.
- Thompson, D. E., Stability of glaciers and ice sheets against flow perturbations, *J. Glaciol.*, 24, 427–441, 1979.
- Thorsteinsson, T., C. F. Raymond, G. H. Gudmundsson, R. B. Bindshadler, P. Vomberger, and I. Joughin, Bed topography and lubrication inferred from surface measurements on fast flowing ice streams, *J. Glaciol.*, in press, 2003.
- van de Wal, R. S. W., and J. Oerlemans, Response of valley glaciers to climate change and kinematic waves: A study with a numerical flow model, *J. Glaciol.*, 41, 142–152, 1995.
- Weber, J. E., Non-temperate glacier flow over wavy sloping ground, *J. Glaciol.*, 46, 2000.
- Weertman, J., Traveling waves on glaciers, in *Symposium at Chamonix 1958-Physics of the Movement of Ice*, IASH Publ., 47, 162–168, 1958.
- Whillans, I. M., and S. J. Johnsen, Longitudinal variations in glacial flow: theory and test using data from the Byrd Station strain network, Antarctica, *J. Glaciol.*, 29, 78–97, 1983.
- Yosida, Z., Internal stress and viscous flow of snow covers on sloping ground surfaces: I. Snow cover on wavy ground of mean inclination α (with English summary), *Low Temp. Sci., Ser. A*, 22, 83–100, 1964.

G. H. Gudmundsson, British Antarctic Survey, High Cross, Madingley Road, Cambridge, CB3 0ET, UK. (ghg@bas.ac.uk)